Notes: Benmelech (RFS, 2009)

Asset Salability and Debt Maturity: Evidence from Nineteenth-Century American Railroads

Contents

1 Big picture

- **Question.** Does the liquidation value, or salability, of collateral assets shape the maturity and amount of debt issued by firms?
- **Empirical setting.** The paper studies nineteenth-century American railroads, an industry with large secured debt, detailed asset-level information, and major variation in track gauge.
- **Core identification idea.** Track gauge determines who can reuse a railroad's tracks and rolling stock. A standard-gauge railroad has many potential buyers for its assets; a non-standard-gauge railroad has fewer buyers.
- **Conceptual distinction.** Salability combines *redeployability* and *liquidity*. Redeployability depends on whether the asset can be used by others. Liquidity depends on whether potential users are financially able to buy it.
- **Main prediction 1.** Higher salability should increase debt capacity, because creditors recover more in default.
- **Main prediction 2.** Higher salability should increase debt maturity, because durable and redeployable assets can support longer-term claims.
- **Main result.** Asset salability is strongly and positively related to debt maturity. Railroads with more salable road assets, more salable rolling stock, and standard-gauge compatibility issue longer-maturity debt.
- **Leverage result.** In the full cross-section, salability is not strongly related to leverage. During the mid-1870s depression, however, more salable railroads raise more debt and are more likely to obtain long-term financing.
- **Key economic mechanism.** When creditors expect high resale value for collateral, they are more willing to lend long and to lend during distress.
- **Why railroads are useful.** The paper observes physical characteristics of assets rather than relying only on accounting tangibility, which can be a poor proxy for liquidation value.
- **Takeaway.** Tangibility is not enough. The identity and financial condition of potential buyers matter for debt contracts.

2 Data and measurement: railroads, gauges, assets, and salability

2.1 Why nineteenth-century railroads are the right laboratory

- Railroads relied heavily on secured debt. Debt contracts were typically backed by tracks, rolling stock, and related property.
- The industry used two main asset categories:
  - **Road and construction:** tracks, ties, roadbed, bridges, and fixed infrastructure.
  - **Rolling stock:** locomotives, freight cars, passenger coaches, and other mobile equipment.
- Track gauge - the horizontal distance between rails - varied widely before standardization.
- Rolling stock could be moved, but only if it fit the buyer's gauge or could be adapted at cost.
- Tracks were site-specific, but nearby railroads operating the same gauge could acquire or lease them to extend a network.

Interpretation. Track gauge creates a physical, historically grounded measure of collateral redeployability.

2.2 Gauge diversity

- The standard gauge was 56.5 inches, or 4 feet 8.5 inches.
- In the pooled sample, 64.6 percent of railroad-year observations use the standard gauge.
- The 60-inch wide gauge is the second most common, with 16.4 percent of observations.
- Standardization rises over time: the standard gauge share increases from 52.2 percent in 1868 to 76.8 percent in 1882.
- The South remains dominated by the 60-inch gauge, while New England and the West are overwhelmingly standard gauge.

Interpretation. Gauge diversity creates variation in the number of potential buyers for assets. Standard gauge assets can be redeployed to many more users.

2.3 Sample and sources

- The main data come from *Poor's Manuals of the Railroads*.
- The panel covers 1868, 1873, 1877, and 1882.
- Railroads enter the sample if they operate at least 100 miles and have enough information for the variables.
- The resulting panel has 390 railroad-year observations representing 221 distinct railroads.
- The author collects assets, equipment, leverage, maturity, profitability, regions of operation, mileage, gauge, and rolling stock composition.

Interpretation. The sample is designed to include medium and large railroads with meaningful debt structures while preserving cross-sectional gauge variation.

2.4 Debt maturity and leverage

Leverage is measured as

$$Leverage_i = \frac{\text{book value of funded debt}_i}{\text{book value of total assets}_i}.$$

Debt maturity is the weighted average maturity of all outstanding debt instruments:

$$Maturity_i = \frac{\sum_{j=1}^{J_i} D_{ij} M_{ij}}{\sum_{j=1}^{J_i} D_{ij}},$$

where D_{ij} is the book value of debt instrument j for railroad i, and M_{ij} is its term to maturity.

Interpretation. The maturity measure captures the whole liability structure rather than only the fraction of short-term debt.

2.5 Key railroad characteristics

- Average leverage is 0.43; median leverage is also 0.43.
- Average debt maturity is 20.1 years; median debt maturity is 20.0 years.
- Railroads are highly tangible: median tangibility is 0.86.
- Road and construction accounts for 70 percent of assets on average.
- Rolling stock accounts for 12 percent of assets on average.
- Freight cars dominate rolling stock: freight cars account for 87 percent of cars on average.

Interpretation. Long-maturity debt is natural because railroad assets are long-lived, but the paper asks which assets can support especially long debt.

2.6 Road salability

Road salability is based on railroads in the same state and same gauge that are not in equity receivership.

For state s, gauge g, and year t, define:

$$mileage^{road}_{s,g,t} = \sum_{b \in B_{s,g,t}} length_b,$$

$$number^{road}_{s,g,t} = \sum_{b \in B_{s,g,t}} I_b,$$

where $B_{s,g,t}$ is the set of solvent railroads operating in state s with gauge g at time t.

For railroad i, the road-salability measures are:

$$mileage^{road}_{i,t} = \sum_s \sum_g \omega_{i,s,g,t} \left(mileage^{road}_{s,g,t} - length_{i,s,g,t} \right),$$

$$number^{road}_{i,t} = \sum_s \sum_g \omega_{i,s,g,t} \left(number^{road}_{s,g,t} - 1 \right),$$

with mileage weights

$$\omega_{i,s,g,t} = \frac{length_{i,s,g,t}}{\sum_s \sum_g length_{i,s,g,t}}.$$

Interpretation. A railroad's road is more salable when many solvent same-gauge railroads operate in the same states.

2.7 Rolling stock salability

Rolling stock is mobile, so potential buyers are railroads nationwide with the same gauge. For gauge g and year t:

$$mileage^{rolling}_{g,t} = \sum_{c \in C_{g,t}} length_c,$$

$$number^{rolling}_{g,t} = \sum_{c \in C_{g,t}} I_c,$$

where $C_{g,t}$ is the set of solvent railroads with gauge g.

For railroad i:

$$mileage^{rolling}_{i,t} = \sum_g \pi_{i,g,t} \left(mileage^{rolling}_{g,t} - length_{i,g,t} \right),$$

$$number^{rolling}_{i,t} = \sum_g \pi_{i,g,t} \left(number^{rolling}_{g,t} - 1 \right),$$

with

$$\pi_{i,g,t} = \frac{length_{i,g,t}}{\sum_g length_{i,g,t}}.$$

Interpretation. Rolling stock salability is broader than road salability because rolling stock can be shipped across regions, but it still requires gauge compatibility.

2.8 Receiverships share

Receivership share captures market illiquidity among potential road buyers:

$$ReceivershipShare_{i,t} = \sum_s \sum_g \omega_{i,s,g,t} \left(\frac{mileage^{road}_{s,g,t,R}}{mileage^{road}_{s,g,t,U}} \right),$$

where R is the set of railroads in receivership and U is the universe of railroads operating in the state-gauge cell.

Interpretation. A higher receivership share means more potential buyers are financially constrained. The same physical asset becomes less liquid when the industry is distressed.

3 Models and empirical specifications

3.1 Incomplete-contracting predictions

The paper tests two predictions:

$$\frac{\partial DebtLevel}{\partial Salability} > 0, \qquad \frac{\partial DebtMaturity}{\partial Salability} > 0.$$

Interpretation. Higher liquidation value strengthens creditors' foreclosure rights. This should support more debt and longer debt maturity.

3.2 Baseline leverage regression

The leverage regressions use:

$$Leverage_{i,t} = \alpha + \beta Salability_{i,t} + \Gamma' X_{i,t} + \delta_t + \varepsilon_{i,t}.$$

Controls include size, tangibility, profitability, freight-share, and year dummies; some specifications add state fixed effects.

Interpretation. The leverage test asks whether salable collateral expands debt capacity in the cross-section.

3.3 Baseline maturity regression

The main maturity regression is:

$$Maturity_{i,t} = \alpha + \beta Salability_{i,t} + \Gamma' X_{i,t} + \delta_t + \varepsilon_{i,t}.$$

The paper estimates the equation separately using five salability measures:

- road salability by mileage;
- road salability by number of buyers;
- rolling stock salability by mileage;
- rolling stock salability by number of buyers;
- receiverships share.

Interpretation. The core hypothesis is $\beta > 0$ for buyer and mileage measures, and $\beta < 0$ for receivership share.

3.4 Economic-significance calculation

The paper converts regression coefficients into percentage changes in long-term debt maturity:

$$\Delta Maturity = \widehat{\beta} \times \Delta Salability.$$

The reported changes are based on:

- one-standard-deviation changes;
- 25th-to-75th percentile changes;
- 10th-to-90th percentile changes.

Interpretation. This shows whether the statistically significant effects are economically large.

3.5 Refinancing during the 1875 depression

The 1875 analysis uses a sample of railroads that raised debt during the depression. It studies:

$$Pr(LongTermBond_i = 1) = \Phi\left(\alpha + \beta Salability_i + \Gamma' X_i\right),$$

$$Pr(CreditLine_i = 1) = \Phi\left(\alpha + \beta Salability_i + \Gamma' X_i\right),$$

and OLS regressions for the amount and maturity of debt financing.

Interpretation. A crisis is a natural stress test. If salable assets are valuable collateral, more salable railroads should be able to issue long-term bonds rather than rely on credit lines.

3.6 Size distribution of potential buyers

The paper also tests whether the absolute size of potential buyers matters:

$$Maturity_{i,t} = \alpha + \beta_1 StateGaugeShare_{i,t} + \beta_2 CountryGaugeShare_{i,t} + \Gamma' X_{i,t} + \varepsilon_{i,t}.$$

It further divides potential buyers into railroads with road length below or above 300 miles.

Interpretation. A railroad can be too large relative to the pool of buyers. Having large solvent buyers is especially valuable.

3.7 Robustness to age, durability, and growth opportunities

The paper controls for:

- age of the railroad;
- steel tracks, as a proxy for rail durability;
- operating efficiency;
- freight earnings;
- area-to-population;
- Herfindahl-Hirschman concentration index;
- waterway competition.

Interpretation. These tests ask whether salability is simply proxying for asset durability, investment timing, market power, or growth opportunities.

4 Identification and empirical design

4.1 What identifies the main effect

- The identifying variation comes from physical compatibility of assets through track gauge and from the number and solvency of potential buyers.
- Gauge was historically influenced by local geography, local trade strategy, engineering decisions, and state rules.
- Once gauge is chosen, it determines the resale market for tracks and rolling stock.
- The main regressions compare railroads with different salability, controlling for size, tangibility, profitability, freight composition, year effects, and sometimes state fixed effects.

Interpretation. The paper treats gauge-based salability as a physical asset-market characteristic rather than a choice made solely for financing reasons.

4.2 Why reverse causality is a concern

- A possible objection is that better-financed railroads may have chosen standard gauge precisely because they had better access to capital markets.
- If so, gauge would be an outcome of financing strength rather than a determinant of debt contracts.
- The paper argues that many gauge choices were historically driven by local strategic and engineering considerations, not by debt maturity.
- State laws and regional conventions also shaped gauge choices.

Interpretation. The instrument-like variation is not perfect, but the historical narrative supports treating gauge as plausibly exogenous to later debt-maturity choices.

4.3 Why financial distress of potential buyers matters

- Shleifer and Vishny's liquidation-value theory emphasizes that buyers may be financially constrained during industry distress.
- Benmelech implements this by subtracting railroads in equity receivership from the buyer pool.
- A high receiverships share lowers liquidity even if many same-gauge railroads exist physically.

Interpretation. This is the paper's main innovation beyond pure redeployability: liquidation value depends on the financial strength of potential buyers.

4.4 What the paper cannot fully prove

- Gauge choice is not randomly assigned.
- The salability proxies are measured with error because actual asset-sale transactions are not observed for all railroads.
- The paper cannot fully separate salability from all local demand, political, or network effects.
- The leverage results are weaker than the maturity results.
- The setting is historically specific: railroads were secured borrowers with physical assets and long-lived infrastructure.

Interpretation. The evidence is strongest for debt maturity and for crisis refinancing. It is less strong for unconditional leverage.

5 Tables and figures

5.1 Tables 1-2 and Figure 1: gauge distribution and sample representativeness

Read: Table 1:

- Standard gauge, 56.5 inches, accounts for 64.6 percent of the pooled sample.
- The 60-inch gauge accounts for 16.4 percent.
- Standard gauge rises from 52.17 percent in 1868 to 76.8 percent in 1882.

Read: Table 2:

- New England and the West are overwhelmingly standard gauge.
- The South is dominated by 60-inch wide gauge: 64.5 percent of southern railroad-year observations use it.
- The East and Midwest have more gauge diversity than New England.

Read: Figure 1:

- The sample's gauge distribution closely tracks the full population mileage distribution.
- This reduces concern that the 100-mile sample cutoff creates major gauge-selection bias.

Economic message. Gauge variation is the paper's quasi-physical source of variation in asset salability.

Table 1
The distribution of track gauge, 1868–1882

Panel A: Distribution of gauge across the pooled data

Gauge (inches)	36	56.5	57	57.25	57.5	58	60	65	66	72	Tota
Frequency	3	252	32	1	14	14	64	1	3	6	390
Percent (%)	0.8	64.6	8.2	0.3	3.6	3.6	16.4	0.3	0.8	1.5	100.

Panel B: Distribution of gauge by year

1868											
Gauge (inches)	36	56.5	57	57.25	57.5	58	60	65	66	72	Tota
Frequency	0	36	2	0	2	4	17	1	2	5	69
Percent (%)	0.0	52.17	2.9	0.0	2.9	5.8	26.6	1.5	2.9	7.3	100.
1873											
Gauge (inches)	36	56.5	57	57.25	57.5	58	60	65	66	72	Tota
Frequency	2	59	7	1	7	5	19	0	1	1	102
Percent (%)	2.0	57.8	6.9	1.0	6.9	4.9	18.6	0.0	1.0	1.0	100.
1877											
Gauge (inches)	36	56.5	57	57.25	57.5	58	60	65	66	72	Tota
Frequency	0	69	7	0	5	5	18	0	0	0	104
Percent (%)	0.0	66.4	6.7	0.0	4.8	4.8	17.3	0.0	0.0	0.0	100.
1882											
Gauge (inches)	36	56.5	57	57.25	57.5	58	60	65	66	72	Tota
Frequency	1	86	16	0	0	0	9	0	0	0	112
Percent (%)	0.9	76.8	14.3	0.0	0.0	0.0	8.0	0.0	0.0	0.0	100.

The sample consists of 390 railroad-year observations in the years: 1868, 1872, 1877, and 1882. The track gaug is the horizontal distance separating the two rails in inches. The "standard gauge" was 4′8.5″ (or 56.5 inches).

Figure 1: Original Table 1: distribution of track gauge, 1868-1882

Table 2
The geographical distribution of the track gauge

New England												
Gauge (inches)	36	56.5	57	57.25	57.5	58	60	62	65	66	72	Total
Frequency	0	53	0	0	0	0	0	0	0	2	0	55
Percent (%)	0.0	96.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.6	0.0	100.0
East												
Gauge (inches)	36	56.5	57	57.25	57.5	58	60	62	65	66	72	Total
Frequency	0	81	14	0	2	4	3	0	0	0	5	109
Percent (%)	0.0	77.3	12.8	0.0	1.8	3.7	2.8	0.0	0.0	0.0	4.6	100.0
South												
Gauge (inches)	36	56.5	57	57.25	57.5	58	60	62	65	66	72	Total
Frequency	1	22	9	0	0	0	60	0	0	0	1	93
Percent (%)	1.1	23.7	9.7	0.0	0.0	0.0	64.5	0.0	0.0	0.0	1.1	100.0
Midwest												
Gauge (inches)	36	56.5	57	57.25	57.5	58	60	62	65	66	72	Total
Frequency	2	112	12	1	13	14	7	0	1	0	2	164
Percent (%)	1.2	68.3	7.3	0.6	7.9	8.5	4.3	0.0	0.6	0.0	1.2	100.0
West												
Gauge (inches)	36	56.5	57	57.25	57.5	58	60	62	65	66	72	Total
Frequency	0	16	0	0	0	0	1	0	0	1	0	18
Percent (%)	0.0	88.9	0.0	0.0	0.0	0.0	5.6	0.0	0.0	5.6	0.0	100.0

This table reports the distribution of the railroads sample across geographical regions and track gauges in the entire (pooled) sample. The geographical categories are in accordance with the railroads geographical groups,

Figure 2: Original Table 2: geographical distribution of track gauge

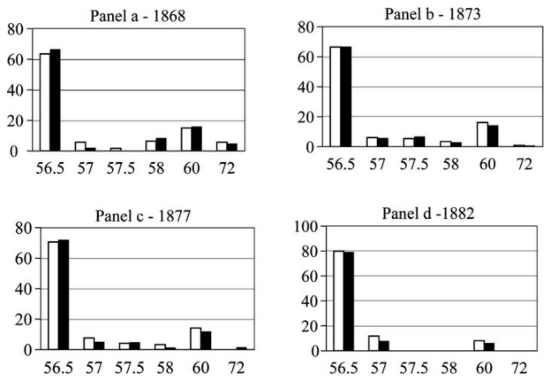

Figure 3: Original Figure 1: sample representativeness

Table 3
Railroads' characteristics

	Mean	25th percentile	Median	75th percentile	Standard deviation	Min	Max
Leverage	0.43	0.30	0.43	0.56	0.19	0.0	1.0
Debt maturity	20.1	14.0	20.0	27.0	9.4	0.0	60.0
Profitability	4.85%	2.03%	4.09%	6.53%	4.30%	−2.77%	50.78
Tangibility	0.84	0.76	0.86	0.95	0.18	0.0	1.0
Road and construction	0.70	0.63	0.76	0.86	0.19	0.01	0.98
Land	0.026	0.002	0.011	0.036	0.037	0.000	0.22
Rolling stock	0.12	0.07	0.10	0.15	0.08	0.02	0.78
Cars	2317.5	427.0	961.0	2150.0	4460.7	0.0	32425.0
Freight	0.87	0.85	0.89	0.93	0.11	0.0	0.99
Passengers	0.043	0.018	0.031	0.051	0.059	0.000	0.87
Locomotives	0.051	0.033	0.046	0.062	0.037	0.005	0.49

This table provides descriptive statistics for the characteristics of the railroads, a sample of 221 America railroads in the years 1868, 1873, 1877, and 1882 (390 firm-year observations). Leverage is measured as th book value of total funded debt divided by the book value of the assets. Debt maturity is the weighted averag of the term-to-maturity of all the debt instruments outstanding. Profitability is earnings before interest expens divided by the book value of total assets. Tangibility is the value of the road and construction, land and rollin stock divided by the book value of assets. Road and construction is the ratio of the book value of road ar construction to total assets. Land is defined as the ratio of the book value of land and real estate to total asset Rolling stock is the ratio of the book value of rolling stock to total assets. Cars is the total number of locomotive passenger coaches, freight cars, baggage mail, express cars, and service cars. Freight is the number of freig cars divided by the total number of cars. Passengers is the number of passenger coaches divided by the tot number of cars. Locomotives is the number of locomotives divided by the total number of cars.

Figure 4: Original Table 3: railroad characteristics

5.2 Table 3: railroad characteristics

Read:

- Average leverage is 0.43.
- Average debt maturity is 20.1 years; the 75th percentile is 27 years.
- Median tangibility is 0.86.
- Road and construction assets are about 70 percent of assets on average.
- Freight cars are about 87 percent of the cars fleet.

Economic message. The industry is highly tangible and debt-heavy, which makes it a clean setting for studying secured debt contracts.

5.3 Table 4: characteristics of salability proxies

Read:

- Road salability has a mean of 1,859.8 miles and a median of 1,162.3 miles.
- Rolling stock salability is much larger because it is measured nationally by gauge.
- Receiverships share has a mean of 5.12 percent and a median of zero.
- Salability rises across debt-maturity groups.
- Salability is not similarly ordered across leverage groups.

Economic message. The univariate patterns already point toward a maturity effect rather than a simple leverage effect.

Table 4
Characteristics of the salability proxies

Panel A: Summary statistics of the salability measures

	Mean	25th percentile	Median	75th percentile	Standard deviation	Min	Max
Road (mileage)	1859.8	440.2	1162.3	3257.0	1857.7	0.0	7649.8
Road (number of buyers)	28.4	5.7	17.7	35.7	34.4	0.0	162
Rolling stock (mileage)	35605.8	6254.0	37262.9	48483	30137.1	0.0	83603.7
Rolling stock (number of buyers)	460.3	63	456	649	407.6	0	1100
Receiverships share (mileage)	5.12%	0.00%	0.0%	8.9%	14.20%	0.00%	100.00%

Panel B: Salability proxies stratified by debt maturity

Maturity ranges	1–8	9–14	15–20	21–27	28–30	31+	Kruskal-Wallis
Road	1265.0	1472.1	1596.3	1781.2	2479.0	3170.6	0.0001
(mileage)	(991.4)	(1026.8)	(1002.2)	(1181.1)	(2359.5)	(3680.8)	
Road	23.93	23.46	27.4	22.7	33.5	50.2	0.0034
(number of buyers)	(12.1)	(15.83)	(17.0)	(15.3)	(31.8)	(30.6)	
Rolling stock	20171.6	29723.5	34323.1	33593.8	48117.8	55142.4	0.0001
(mileage)	(7226)	(19535)	(37334.9)	(37218.9)	(48289.1)	65310.9	
Rolling stock	265.4	366.0	452.0	437.0	622.4	710.8	0.0001
(number of buyers)	(105)	(217)	(456)	(456)	(649)	(874.5)	
Receiverships share	5.23%	4.26%	3.56%	3.58%	2.03%	1.77%	N/A[a]
(mileage)	(0.00%)	(0.00%)	(0.00%)	(0.00%)	(0.00%)	(0.00%)	

Panel C: Salability proxies stratified by leverage

Leverage ranges	0–0.18	0.19–0.29	0.30–0.43	0.438–0.55	0.56–0.66	0.67+	Kruskal-Wallis
Road	1797.8	1805.0	1661.0	2203.2	2039.7	1541.9	0.46
(mileage)	(1236.6)	(1104.7)	(999.3)	(1389.4)	(1181.1)	(1069.7)	
Road	32.6	30.1	26.2	28.9	34.7	20.7	0.34
(number of buyers)	(22.0)	(19.6)	(13.2)	(21.0)	(18.0)	(14.0)	
Rolling stock	31376.9	37618.3	34802.3	37760.4	39173.4	33390.1	0.75
(mileage)	(19658.5)	(37352.4)	(36692.9)	(37405.9)	(37376.9)	(37171.9	
Rolling stock	395.2	474.0	453.5	499.5	514.5	417.0	0.84
(number of buyers)	(217)	(456)	(456)	(649)	(456)	(456)	
Receiverships share	4.90%	2.19%	3.62%	3.47%	3.50%	3.19%	0.32
(mileage)	(0.00%)	(0.00%)	(0.00%)	(0.00%)	(0.001%)	(0.00%)	

This table provides descriptive statistics for the characteristics of the salability proxies. Panel A reports summary statistics for the proxies of road salability, rolling stock salability, and receiverships share. I use both total mileage and number of railroads to calculate the proxies of road and rolling stock salability. Panel B reports means (medians) of the proxies of road salability, rolling stock salability, and receiverships share stratified by debt maturity. Panel C reports means (medians) of the proxies of road salability, rolling stock salability, and receiverships share stratified by leverage. The Kruskal-Wallis *p*-value gives the significance of a test of whether a variable is not distributed identically across debt ma-

Figure 5: Original Table 4: characteristics of salability proxies

5.4 Table 5: asset salability and leverage

Read:

- The dependent variable is leverage.
- Salability coefficients are generally small and statistically insignificant.
- Profitability is negatively related to leverage in most specifications.
- The evidence does not show a strong cross-sectional relation between salability and leverage.

Economic message. Salability does not clearly explain how much debt railroads carry in the full cross-section, but it matters strongly for maturity and for borrowing during distress.

Table 5
Asset salability and leverage

	Dependent variable = leverage					
	Model 1	Model 2	Model 3	Model 4	Model 5	Model
Size	−0.002	0.001	0.001	−0.003	0.001	−0.02
	(−0.12)	(0.01)	(0.01)	(−0.02)	(0.06)	(−1.7
Tangibility	0.113	0.110	0.113	0.115	0.127	0.010
	(1.36)	(1.31)	(1.42)	(1.44)	(1.64)	(1.03)
Profitability	−0.783**	−0.769**	−0.759**	−0.768**	−0.833**	−0.74
	(−2.54)	(−2.44)	(−2.35)	(−2.38)	(−2.54)	(−1.8
Freight	0.101	0.107	0.103	0.104	0.114	0.080
	(0.94)	(0.94)	(1.11)	(1.13)	(1.29)	(0.69)
Road salability	−0.005					−0.00
(mileage)	(−1.25)					(−0.3
Road salability		−0.004				
(number of buyers)		(−0.94)				
Rolling stock salability			−0.007			
(mileage)			(−1.08)			
Rolling stock salability				−0.000		
(number of buyers)				(−0.41)		
Receiverships share					−0.105	
(mileage)					(−1.46)	
1873 dummy	0.052	0.056*	0.058	0.056	0.048	0.054
	(1.63)	(1.76)	(1.42)	(1.38)	(1.20)	(1.23)
1877 dummy	0.077**	0.083**	0.081**	0.080*	0.075*	0.072
	(2.45)	(2.53)	(2.00)	(1.93)	(1.70)	(1.68)
1882 dummy	0.061*	0.072**	0.070*	0.068	0.054	0.063
	(1.89)	(2.04)	(1.86)	(1.57)	(1.55)	(1.60)
State-fixed effects	No	No	No	No	No	Yes
Adjusted R^2	0.06	0.06	0.06	0.06	0.08	0.13
Observations	382	382	382	382	382	382

The dependent variable in the regressions is the book value of total funded debt divided by the book value the assets. Size is the log of the book value of assets. Tangibility is the value of the road and construction, lan and rolling stock divided by the book value of assets. Profitability is earnings before interest expenses divid by the book value of total assets. Freight is the number of freight cars divided by the total number of cars. Ro salability is defined as the mileage-weighted average of the state salability index corresponding to the states of t railroad's line, where the state salability index is calculated using (i) statewide track mileage for each gauge, a (ii) the number of railroads in the state for each gauge. Rolling stock salability is defined as the mileage-weight average of the gauge salability index corresponding to the railroad's gauge, where as before the state salabili index is calculated using (i) statewide track mileage for each gauge, and (ii) the number of railroads in the sta for each gauge. The salability proxies that are calculated using mileage are in logarithm terms. Receivershi share is the tracks mileage share of potential buyers of roads that are in receiverships. All regressions include intercept (not reported) and years dummies (year 1868 omitted). t-statistics are calculated using robust standa errors that are clustered by state and reported in parentheses.

Figure 6: Original Table 5: asset salability and leverage

5.5 Table 6 and Table 7: asset salability and debt maturity

Read: Table 6:

- Road salability by mileage is positive and significant in model 1, with coefficient 0.262.
- Road salability by number of buyers is positive and significant in model 2, with coefficient 0.023.
- Rolling stock salability by mileage is positive and significant in model 3, with coefficient 0.633.
- Rolling stock salability by number of buyers is positive and significant in model 4, with coefficient 0.004.
- Receiverships share is negative and significant in model 5, with coefficient -11.14.
- With state fixed effects, the positive salability effects remain strong for road and rolling-stock measures.

Read: Table 7:

- A one-standard-deviation increase in road salability by number of buyers raises maturity by 8.19 percent.
- A one-standard-deviation increase in rolling-stock salability by mileage raises maturity by 6.15 percent.
- A one-standard-deviation increase in receiverships share lowers maturity by 7.87 percent.
- Moving from the 10th to the 90th percentile of rolling-stock buyer salability raises maturity by 16.30 percent.

Economic message. These are the central results. More salable collateral supports longer debt, while illiquidity among potential buyers shortens debt maturity.

Table 6
Asset salability and debt maturity

	Dependent variable = Debt maturity									
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10
Size	2.37***	2.28***	2.24***	2.22***	2.25***	1.31***	1.19**	1.17**	1.22**	1.20**
	(5.35)	(5.00)	(4.80)	(4.75)	(4.98)	(2.75)	(2.54)	(2.30)	(2.42)	(2.28)
Tangibility	7.75***	8.00***	7.87***	8.15***	7.97***	4.07*	4.43*	4.04*	4.12*	3.91
	(2.61)	(2.73)	(2.61)	(2.65)	(2.69)	(1.72)	(1.91)	(1.72)	(1.82)	(1.54)
Profitability	−26.86**	−27.60**	−29.04**	−28.82***	−29.00**	−28.87*	−26.36*	−29.46**	−28.49**	−29.57**
	(−1.97)	(−2.13)	(−2.20)	(−2.23)	(−2.15)	(−1.95)	(−1.89)	(−2.16)	(−2.10)	(−2.02)
Freight	13.25***	13.31***	13.70***	13.61***	13.45***	6.96*	6.67*	7.24**	7.42**	7.66**
	(4.17)	(4.25)	(4.02)	(4.17)	(3.98)	(1.91)	(1.87)	(2.06)	(2.12)	(2.17)
Road salability (mileage)	0.262**					0.380**				
	(2.20)					(2.37)				
Road salability (number of buyers)		0.023***					0.049***			
		(2.66)					(3.20)			
Rolling stock salability (mileage)			0.633***					0.749***		
			(2.61)					(2.62)		
Rolling stock salability (number of buyers)				0.004***					0.003*	
				(2.62)					(1.92)	
Receiverships share (mileage)					−11.14**					−4.83
					(−4.73)					(−1.51)
1873 dummy	3.56***	3.37***	3.19***	3.09**	3.59***	3.84***	3.40***	3.33***	3.33***	3.71***
	(3.26)	(2.94)	(2.78)	(2.67)	(3.09)	(3.88)	(3.19)	(3.14)	(3.05)	(3.42)
1877 dummy	2.14	1.76	1.74	1.12	2.70*	2.26*	1.24	1.91	1.43	2.58*
	(1.67)	(1.33)	(1.32)	(0.84)	(1.93)	(1.78)	(1.02)	(1.48)	(1.10)	(1.89)
1882 dummy	6.04***	5.29***	5.05***	3.69**	6.36***	6.32***	4.51**	5.31***	4.34***	6.54***
	(3.68)	(3.00)	(3.01)	(2.02)	(3.70)	(3.96)	(2.72)	(3.09)	(2.11)	(3.81)
State-fixed effects	No	No	No	No	No	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.20	0.20	0.20	0.21	0.22	0.29	0.29	0.28	0.28	0.27
Observations	379	379	379	379	379	379	379	379	379	379

The dependent variable in the regressions is the weighted average of the term-to-maturity of all the debt instruments outstanding. Size is the log of the book value of assets. Tangibility is the value of the road and construction, land and rolling stock divided by the book value of assets. Profitability is earnings before interest expenses divided by the book value of total assets. Freight is the number of freight cars divided by the total number of cars. Road salability is defined as the mileage-weighted average of the state salability index corresponding to the states of the railroad's line, where the state salability index is calculated using (i) statewide track mileage for each gauge, and (ii) the number of railroads in the state for each gauge. Rolling stock salability is defined as the mileage-weighted average of the gauge salability index corresponding to the railroad's gauge, where as before the state salability index is calculated using (i) statewide track mileage for each gauge, and (ii) the number of railroads in the state for each gauge. The salability proxies that are calculated using mileage are in logarithm terms. Receiverships share is the tracks mileage share of potential buyers of roads that are in receiverships. All regressions include an intercept (not reported) and years dummies (year 1868 omitted). t-statistics are calculated using robust standard errors that are clustered by state and reported in parentheses.
*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Figure 7: Original Table 6: asset salability and debt maturity

Table 7
Economic significance of the determinants of debt maturity

	Long-term debt		
	Standard deviation	25th–75th percentile	10th–90th percentile
Size	6.39%	9.08%	16.35%
Tangibility	3.65%	3.85%	8.26%
Profitability	−6.17%	−6.45%	12.07%
Freight	3.80%	2.78%	6.37%
Road salability (mileage)	6.28%	14.86%	16.09%
Road salability (number of buyers)	8.19%	7.78%	15.6%
Rolling stock salability (mileage)	6.15%	10.17%	13.69%
Rolling stock salability (number of buyers)	3.30%	6.61%	16.30%
Receiverships share (mileage)	−7.87%	−4.93%	−9.87%

Predicted changes in the dependent variables as each explanatory variable varies (i) by one standard deviation, (ii) from the 25th percentile to the 75th percentile, (iii) from the 10th percentile to the 90th percentile. The results are computed using the specifications in Table 6.

Figure 8: Original Table 7: economic significance of determinants of debt maturity

5.6 Table 8: size distribution of railroads and debt maturity

Read:

- State-gauge share and country-gauge share are negatively related to maturity in several specifications.
- Having potential buyers larger than 300 miles is positively related to maturity.
- The results suggest that salability is not just the number of similar railroads; the size and financial capacity of potential buyers also matter.

Economic message. A large railroad needs sufficiently large buyers. A thin buyer pool can reduce salability even when same-gauge buyers exist.

Table 8
The size distribution of railroads and debt maturity

	Dependent variable = Debt maturity					
	Model 1	Model 2	Model 3	Model 4	Model 5	Model
Mileage	3.08***	2.88***	1.92**	1.68**	1.40***	1.34**
	(4.22)	(4.05)	(2.50)	(2.26)	(2.76)	(2.53)
Tangibility	6.15**	6.20**	3.53	3.25	4.18*	3.40
	(2.01)	(1.97)	(1.47)	(1.30)	(1.81)	(1.32)
Profitability	−28.58*	−29.26**	−32.44**	−32.29**	−30.72**	−43.78
	(−1.93)	(−2.03)	(−2.14)	(−2.22)	(−2.32)	(−2.46
Freight	14.13***	14.54***	7.11*	7.36**	10.84***	11.37**
	(3.89)	(3.82)	(1.95)	(2.07)	(3.56)	(3.67)
State-gauge share	−1.92		−3.19**			
	(−1.63)		(−2.31)			
Country-gauge share		−5.06***		−7.20***		
		(−3.39)		(−2.90)		
No. of railroads ≤ 300					0.245***	
					(3.52)	
No. of railroads > 300						1.49**
						(2.16)
Year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State-fixed effects	No	No	Yes	Yes	Yes	Yes
Adjusted R^2	0.19	0.18	0.29	0.28	0.26	0.25
Observations	379	379	379	379	379	379

The dependent variable in the regressions is the weighted average of the term-to-maturity of all the del instruments outstanding. Mileage is the log of the total mileage owned by the railroad. Tangibility is the value road and equipment divided by the book value of assets. Profitability is earnings before interest expenses by th book value of total assets. Freight is the number of freight cars divided by the total number of cars. State-gaug share is the ratio between the railroad's mileage and the corresponding mileage of railroads with similar gaug that operate in the same states and are not in receiverships. Country-gauge share is the ratio between the railroad mileage and the corresponding mileage of railroads with similar gauge in the country that are not in receivership Number of railroads ≤300 is the number of railroads with similar gauge that operate in the same states and a not in receiverships with total length smaller than or equal to 300 miles. Number of railroads >300 is the numb of railroads with similar gauge that operate in the same states and are not in receiverships, with total leng larger than 300 miles. All regressions include an intercept (not reported). t-statistics are calculated using robu standard errors that are clustered by state and reported in parentheses.
*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Figure 9: Original Table 8: size distribution of railroads and debt maturity

5.7 Table 9: evidence from the 1875 refinancing episode

Read:

- Panel A shows that higher road and rolling-stock salability increases the probability of long-term bond financing.
- The same salability measures reduce the probability of credit-line financing.
- Receiverships share is negatively related to long-term bond financing and positively related to credit-line financing.
- Panel B shows that salability raises both the amount and maturity of debt raised in 1875.
- Receiverships share lowers the maturity of debt raised.

Economic message. During the mid-1870s depression, salable assets helped railroads borrow more and borrow long. This is the clearest debt-capacity evidence in the paper.

Table 9
Asset salability and debt refinancing: Evidence from 1875

	Panel A: Asset salability and the probability of long-term/short-term financing									
	Probability of long-term bond financing					Probability of credit line financing				
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10
Road salability (mileage)	0.044**					−0.058***				
	(2.26)					(−3.07)				
Road salability (number of buyers)		0.008*					−0.001			
		(1.81)					(−0.53)			
Rolling stock salability (mileage)			0.102***					−0.101***		
			(3.38)					(−4.70)		
Rolling stock salability (number of buyers)				0.001***					−0.001***	
				(3.44)					(−4.90)	
Receiverships share (mileage)					−0.607**					0.426***
					(−2.27)					(4.88)
Pseudo R^2	0.16	0.16	0.18	0.17	0.13	0.44	0.28	0.26	0.27	0.20
Observations	102	102	102	102	102	102	102	102	102	102
	Panel B: Asset salability and the amount and maturity of debt financing									
Dependent Variable	Amount	Amount	Amount	Amount	Amount	Maturity	Maturity	Maturity	Maturity	Maturity
Road salability (mileage)	0.055***					2.362***				
	(2.70)					(2.84)				
Road salability (number of buyers)		0.010*					0.503**			
		(1.76)					(2.25)			
Rolling stock salability (mileage)			0.208***					3.505		
			(3.02)					(1.47)		
Rolling stock salability (number of buyers)				0.001***					0.039**	
				(3.41)					(2.15)	
Receiverships share (mileage)					0.670					−17.39***
					(0.99)					(−3.40)
Adjusted R^2	0.40	0.40	0.39	0.42	0.40	0.62	0.60	0.44	0.53	0.37
Observations	53	53	53	53	53	53	53	53	53	53

The dependent variable in Panel A is the probability of long-term bond financing (Models 1–5), or credit line financing (Models 6–10). The dependent variable in Panel B is either the natural logarithm of the amount of debt raised or the maturity of the debt. All regressions control for size, tangibility, profitability, and the number of freight cars divided by the total number of cars. Road salability is defined as the mileage-weighted average of the state salability index corresponding to the states of the railroad's line, where the state salability index is calculated using (i) statewide track mileage for each gauge, and (ii) the number of railroads in the state for each gauge. Rolling stock salability is defined as the mileage-weighted average of the gauge salability index corresponding to the railroad's gauge, where as before the state salability index is calculated using (i) statewide track mileage for each gauge, and (ii) the number of railroads in the state for each gauge. The salability proxies that are calculated using mileage are in logarithm terms. Receiverships share is the tracks mileage share of potential buyers of roads that are in receiverships. All regressions include an intercept (not reported). Regressions in Panel A are estimated using a probit model, and regressions in Panel B are estimated using OLS. t-statistics are calculated using robust standard errors that are clustered by state and reported in parentheses.
*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Figure 10: Original Table 9: asset salability and debt refinancing, 1875

5.8 Tables 10-12: robustness and alternative explanations

Read: Table 10:

- Salability remains positively related to maturity after controlling for age and steel tracks.
- Steel tracks are positively related to maturity, but salability effects do not disappear.

Read: Table 11:

- Salability is not significantly related to profitability.
- Profitability means and medians do not show strong monotonic patterns across salability groups.
- Profitability volatility also does not vary strongly across salability groups.

Read: Table 12:

- Rolling-stock salability remains positively related to debt maturity after controlling for efficiency, freight earnings, area-to-population, HHI, and waterway competition.
- This weakens the interpretation that salability is merely proxying for growth opportunities.

Economic message. The robustness tables support the interpretation that salability itself, rather than profitability, durability, or growth opportunities, drives the maturity result.

Table 10
Railroad's age, asset durability, and debt maturity

	Dependent variable = Debt maturity							
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Freight	11.73***	9.76**	11.23***	9.42**	17.08***	6.47*	6.12*	13.62***
	(2.66)	(2.19)	(2.61)	(2.09)	(5.31)	(1.80)	(1.73)	(3.41)
Road salability	0.266					0.393**		
(mileage)	(1.12)					(2.41)		
Road salability		0.058***					0.050***	
(number of buyers)		(4.34)					(4.93)	
Rolling stock salability			1.17**					
(mileage)			(2.16)					
Rolling stock salability				0.005***				
(number of buyers)				(3.82)				
Receiverships share					−7.81**			−9.98***
(mileage)					(−2.18)		(−3.70)	
Age	−0.057	−0.064	−0.067	−0.072	−10.108			
	(−0.73)	(−0.87)	(−0.89)	(−0.97)	(−1.67)			
Steel						5.25**	5.25**	5.37**
						(2.42)	(2.36)	(2.78)
State-fixed effects	Yes	Yes	Yes	Yes	No	Yes	Yes	No
Year-fixed effects	No	No	No	No	No	Yes	Yes	Yes
Adjusted R^2	0.21	0.23	0.22	0.24	0.17	0.27	0.28	0.21
Observations	326	326	326	326	326	379	379	379

The dependent variable in the regressions is the weighted average of the term-to-maturity of all the debt instruments outstanding. Road salability is defined as the mileage-weighted average of the state salability index corresponding to the states of the railroad's line, in which the state salability index is calculated using (i) statewide track mileage for each gauge, and (ii) the number of railroads in the state for each gauge. Rolling stock salability is defined as the mileage-weighted average of the gauge salability index corresponding to the railroad's gauge, where as before the state salability index is calculated using (i) statewide track mileage for each gauge, and (ii) the number of railroads in the state for each gauge. The salability proxies that are calculated using mileage are in logarithm terms. Receiverships share is the tracks mileage share of potential buyers of roads that are in receiverships. Age is the difference between the actual year and the railroad's charter year. Steel is the ratio of the length of steel tracks to the total mileage of the railroad. Regressions include intercepts, and controls for the proportion of freight cars, size, tangibility, and profitability. For brevity, only the coefficient on the freight salability proxies, age, and steel are reported. t-statistics are calculated using robust standard errors that are clustered by state and reported in parentheses.
*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Figure 11: Original Table 10: age, durability, and maturity

Table 12
Debt maturity and alternative measures of growth opportunities

	Dependent variable = Debt maturity				
	Model 1	Model 2	Model 3	Model 4	Model 5
Freight	5.70*	9.64**	11.28	10.37***	17.45***
	(1.78)	(2.57)	(3.42)	(3.26)	(5.48)
Rolling stock salability	0.756***	0.409*	1.29***	1.20***	1.05***
(mileage)	(2.70)	(1.72)	(3.15)	(3.24)	(3.56)
Efficiency	−10.34***				
	(−2.58)				
Freight earnings		−12.26**			
		(−2.10)			
Area-to-population			−44.55		
			(−1.42)		
HHI				−7.99	
				(−0.65)	
Waterways					0.097
					(0.88)
State-fixed effects	Yes	Yes	Yes	Yes	No
Year-fixed effects	Yes	Yes	No	No	No
Adjusted R^2	0.31	0.28	0.26	0.26	0.20
Observations	371	318	376	376	376

The dependent variable in the regressions is the weighted average of the term-to-maturity of all the debt instruments outstanding. Size is the log of the book value of assets. Tangibility is the value of the road and construction, land, and rolling stock divided by the book value of assets. Profitability is earnings before interest expenses divided by the book value of total assets. Freight is the number of freight cars divided by the total number of cars. Rolling stock salability is defined as the mileage-weighted average of the gauge salability index corresponding to the railroad's gauge, where the state salability index is calculated using statewide track mileage for each gauge. The salability proxy is in logarithm terms. Efficiency is the ratio of operating expenses to revenue. Freight earning is defined as earnings from freight divided by total earnings. Area-to-population is the ratio of the state's area in squared miles to the population size. HHI is the Herfindahl-Hirschman concentration index of railroads at the state-year level. Waterways is defined as the number of navigable streams and rivers, and canals within a state. Regressions include intercepts and controls for size tangibility and profitability that are not reported for brevity. All regressions include an intercept (not reported). t-statistics are calculated using robust standard errors that are clustered by state and reported in parentheses.
*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Figure 12: Original Table 12: growth opportunities and maturity

Table 11
Asset salability and profitability

	Model 1	Model 2	Model 3	Model 4	Model 5
	Panel A: Profitability and asset salability Dependent variable = Profitability				
Road (mileage)	0.001 (0.06)				
Road (number of buyers)		−3.0e−04 (−0.26)			
Rolling stock (mileage)			0.010 (0.46)		
Rolling stock (number of buyers)				3.28e−06 (0.37)	
Receiverships share (mileage)					−0.016 (−1.58)
Adjusted R^2	0.18	0.18	0.18	0.18	0.18
	Panel B: Mean and median profitability stratified by salability				
	Low salability 1	2	3	4	High salability 5
Road (mileage)	4.20% (3.60%)	4.09% (4.05%)	4.42% (4.44%)	5.01% (4.51)%	4.36% (4.02%)
Road (number of buyers)	4.09% (3.48%)	4.34% (4.44%)	3.89% (3.42%)	4.93% (4.70)%	4.76% (4.49%)
Rolling stock (mileage)	4.27% (3.70%)	4.18% (3.55%)	4.34% (4.02%)	5.24% (5.13%)	4.12% (4.20%)
Rolling stock (number of buyers)	4.32% (3.75%)	4.47% (4.14%)	4.62% (4.40%)	4.21% (4.25%)	4.41% (4.31%)
	Panel C: Profitability volatility stratified by salability				
	Low salability 1	2	3	4	High salability 5
Road (mileage)	2.77%	2.36%	2.31%	2.79%	2.70%
	Bartlett's test for equal variances: $\chi^2(4) = 4.47$ (p-value = 0.346)				
Road (number of buyers)	2.76%	2.38%	2.33%	2.87%	2.50%
	Bartlett's test for equal variances: $\chi^2(4) = 4.57$ (p-value = 0.334)				
Rolling stock (mileage)	2.74%	2.56%	2.54%	2.69%	2.35%
	Bartlett's test for equal variances: $\chi^2(4) = 1.91$ (p-value = 0.753)				
Rolling stock (number of buyers)	2.57%	2.77%	2.61%	2.44%	2.37%
	Bartlett's test for equal variances: $\chi^2(4) = 1.30$ (p-value = 0.728)				

This table examines the relation between profitability and salability. Panel A reports the results from regressing profitability on the salability measures. All regressions include an intercept, year- and state-fixed effects, and a vector of control variables (size, freight, the railroad's age and Herfindahl-Hirschman concentration index of railroads at the state-year level), which are not reported for brevity. t-statistics are calculated using robust standard errors that are clustered by state and reported in parentheses. Panel B reports means (medians) of profitability stratified by different salability measures. I use both total mileage and number of railroads to calculate the proxies. Panel C reports the cross-sectional standard deviation of profitability stratified by proxies of road salability and rolling stock salability. Profitability is earnings before interest expenses divided by the book value of total assets. The Bartlett's p-value gives the significance of a test whether the subsamples have equal variances. Low p-values indicate that the null hypothesis of equal variances is rejected.

Figure 13: Original Table 11: asset salability and profitability

6 Short summary

- The paper tests incomplete-contracting theories of liquidation value using nineteenth-century American railroads.
- The key empirical innovation is measuring asset salability with track-gauge compatibility and the number of solvent potential buyers.
- Standard-gauge rolling stock is more redeployable because many other railroads can use it.
- Road assets are less mobile, so potential buyers are railroads in the same states and same gauge.
- Receiverships share captures market illiquidity: potential buyers in financial distress cannot easily buy used capital.
- The sample has 390 railroad-year observations from 1868, 1873, 1877, and 1882.
- Railroads are highly tangible and typically have long debt maturities.
- Salability is positively associated with debt maturity but not strongly associated with leverage in the full panel.
- More salable assets allow railroads to issue longer maturity debt.
- A higher share of potential buyers in receivership shortens maturity.
- During the 1875 depression, more salable railroads are more likely to issue long-term bonds and less likely to rely on credit lines.
- The results survive controls for size, tangibility, profitability, freight share, state and year effects, railroad age, steel rails, competition, and growth-opportunity proxies.
- The paper's main message is that collateral value depends on who can buy the asset in liquidation and whether those buyers have the financial strength to do so.

7 Conclusion

7.1 Core conclusion

Benmelech shows that asset salability is a key determinant of debt maturity. Railroads whose assets could be sold to more solvent same-gauge buyers issued longer maturity debt. This supports incomplete-contracting theories in which liquidation value affects debt contracting.

7.2 Economic intuition

Creditors care about what collateral is worth after default. A locomotive or track segment may be physically tangible, but its liquidation value depends on whether another railroad can use it. Gauge compatibility expands the set of possible users. Receiverships shrink that set by making potential buyers financially constrained. When assets are easy to sell to solvent buyers, creditors can accept longer maturities.

7.3 Takeaway

The paper's lasting lesson is that "tangible assets" are not the same as "salable assets." Debt contracts depend on market thickness and financial health in the collateral resale market. This insight matters well beyond railroads: collateral values are high only when assets can be redeployed to buyers who can actually pay for them.